

Statistical Methods in Chemistry

Block 2

Repeated Measurements and Statistical Summaries

Student handout

Name: _____

Class: _____

Date: _____

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1 Why repeated measurements matter

One reading is rarely enough in chemistry. A single titre, mass, absorbance, or pH reading may be affected by random variation. Repeating the measurement gives evidence about the reliability of the result.

The role of repetition

Repeated measurements help answer two questions:

- 1) What is the best estimate of the value? This is usually described by the **mean**.
- 2) How much do the readings scatter? This is described by the **range**, **standard deviation**, or **relative standard deviation**.

There is a difference between repeated instrument readings and independent replicates. Measuring the same solution in a colorimeter three times tests short-term instrument repeatability. Preparing three separate standard solutions and measuring each one tests more of the whole method. The second type is usually stronger evidence.

Do not confuse repetition with independence

If all trials use the same wrongly prepared standard solution, repeating the titration will not remove that systematic error. Repetition mainly helps with random scatter. It does not automatically make a biased method accurate.

Quick exercise: Repeated readings or independent replicates?

Classify each case as mainly **repeated readings** or **independent replicates**.

- a) A student places the same cuvette into the colorimeter three times and records three absorbance readings.
 - b) A student prepares three separate copper sulfate standards of the same concentration and measures each one.
 - c) A student titrates three separate aliquots of vinegar using fresh portions of the same standardized NaOH solution.
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2 Mean, median, and range

2.1 The mean

The **arithmetic mean** is the usual best estimate when repeated measurements are affected mainly by random error.

Mean

For measurements $x_1, x_2, \dots, x_n$, the mean is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

The symbol $\bar{x}$ is read as “x-bar”.

2.2 The median

The **median** is the middle value after the data have been arranged in order. It is less affected by one extreme value than the mean. This makes it useful when checking whether a suspicious reading is pulling

the mean away from the main cluster.

2.3 The range

The **range** is the largest value minus the smallest value:

$$\text{range} = x_{\max} - x_{\min}.$$

It is quick, but it uses only two values. It is therefore less informative than standard deviation.

Worked example: Mean, median, and range for titres

A student obtains four concordant titres:

$$24.58 \text{ cm}^3, \quad 24.61 \text{ cm}^3, \quad 24.60 \text{ cm}^3, \quad 24.59 \text{ cm}^3.$$

The mean is

$$\bar{x} = \frac{24.58 + 24.61 + 24.60 + 24.59}{4} = 24.595 \text{ cm}^3.$$

The ordered values are 24.58, 24.59, 24.60, 24.61, so the median is

$$\frac{24.59 + 24.60}{2} = 24.595 \text{ cm}^3.$$

The range is

$$24.61 - 24.58 = 0.03 \text{ cm}^3.$$

Quick exercise: Basic summary statistics

A colorimeter gives the following absorbance readings for the same sample:

$$0.204, \quad 0.207, \quad 0.203, \quad 0.206, \quad 0.205.$$

Calculate the mean, median, and range.

3 Standard deviation: measuring scatter

The mean tells us the central estimate. It does not tell us how tightly the readings are grouped. For that, chemists commonly use the **sample standard deviation**, written as s .

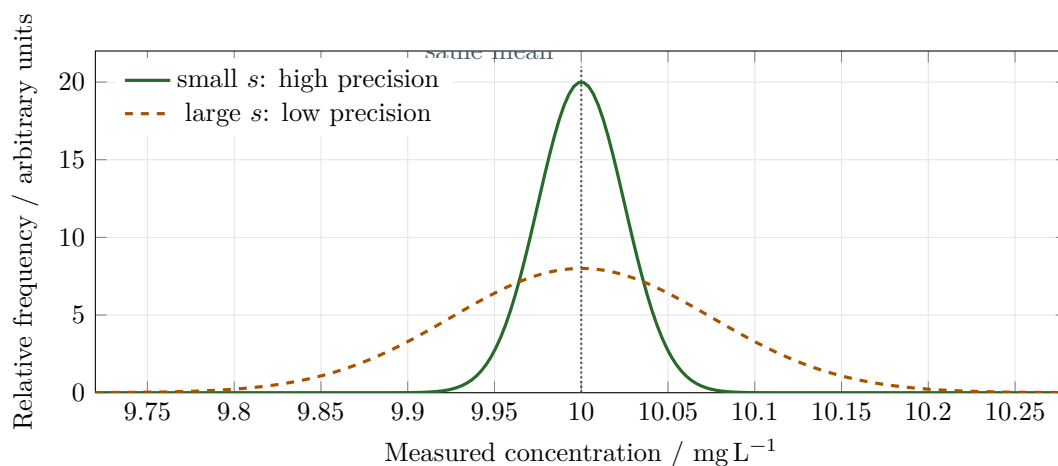
Sample standard deviation

For n repeated measurements, the sample standard deviation is

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}.$$

The denominator is $n - 1$, not n , because we are usually using a sample of measurements to estimate the scatter of a wider measurement process.

A small standard deviation means the readings are close together. A large standard deviation means the readings are more spread out. Standard deviation has the same unit as the measurement.



Worked example: Calculating sample standard deviation

For the titres

$$24.58, \quad 24.61, \quad 24.60, \quad 24.59 \text{ cm}^3,$$

we already found

$$\bar{x} = 24.595 \text{ cm}^3.$$

The deviations are

$$-0.015, \quad +0.015, \quad +0.005, \quad -0.005 \text{ cm}^3.$$

Then

$$s = \sqrt{\frac{(-0.015)^2 + (0.015)^2 + (0.005)^2 + (-0.005)^2}{4 - 1}} = 0.0129 \text{ cm}^3.$$

Rounded sensibly, $s \approx 0.013 \text{ cm}^3$.

Quick exercise: Standard deviation by calculation

Use the absorbance data from the previous exercise:

$$0.204, \quad 0.207, \quad 0.203, \quad 0.206, \quad 0.205.$$

Calculate the sample standard deviation. State what the value means in words.

4 Relative standard deviation

Standard deviation is measured in the same units as the original quantity. This can make it hard to compare precision across different scales. The **relative standard deviation** solves this by expressing scatter as a percentage of the mean.

Relative standard deviation

$$\text{RSD}\% = \frac{s}{\bar{x}} \times 100\%.$$

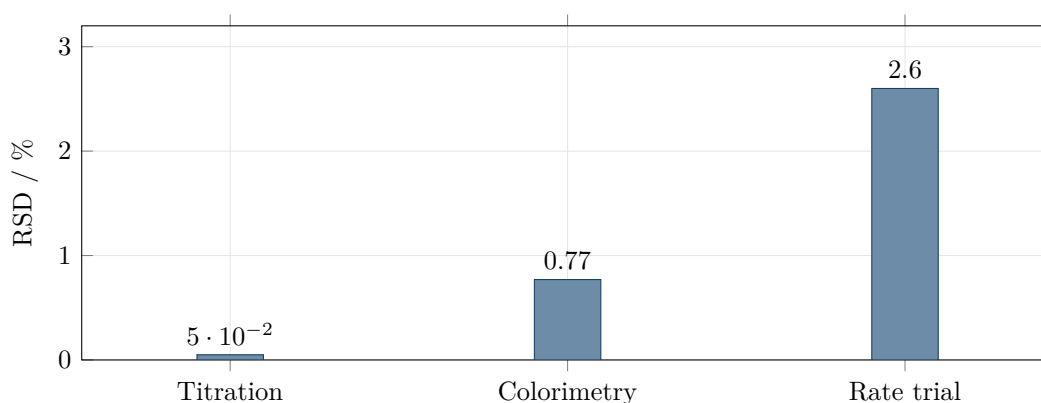
RSD is also called the coefficient of variation. In analytical chemistry, it is often used to compare method precision.

Worked example: RSD for titres

For the titre data, $\bar{x} = 24.595 \text{ cm}^3$ and $s = 0.0129 \text{ cm}^3$.

$$\text{RSD}\% = \frac{0.0129}{24.595} \times 100\% = 0.052\%.$$

This is a very small RSD, which means the repeated titres are highly precise.



The bar chart above compares precision using RSD rather than raw standard deviation. The message is not that titrations are always better than rate experiments. The message is that RSD allows precision to be compared on a common percentage scale.

Quick exercise: RSD for repeated masses

A group measures the mass of product from five repeated trials:

12.41 g, 12.39 g, 12.44 g, 12.40 g, 12.43 g.

Calculate the mean, sample standard deviation, and RSD%.

5 Confidence intervals: uncertainty in the mean

Standard deviation describes the scatter of individual readings. But often we want to know how uncertain the **mean** is. This is where the confidence interval is useful.

95% confidence interval for a mean

For a small set of repeated measurements,

$$\bar{x} \pm t \frac{s}{\sqrt{n}}.$$

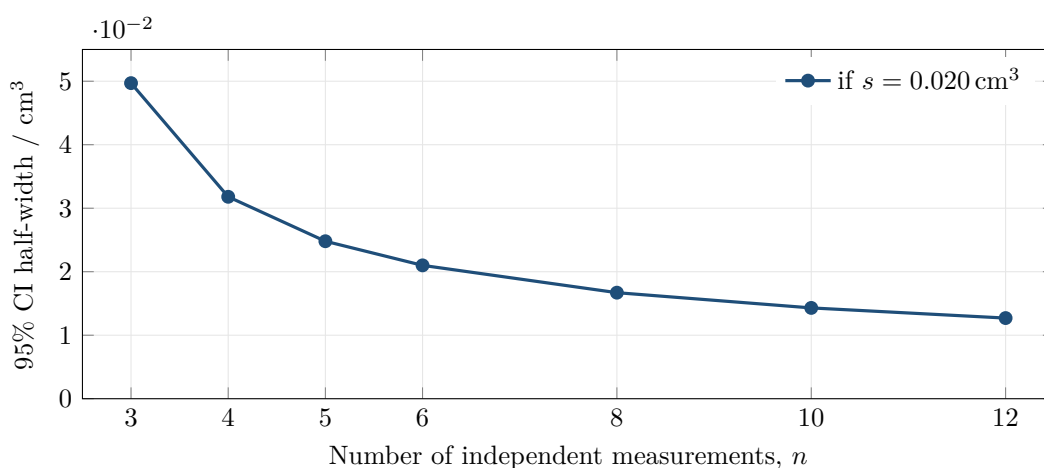
Here s is the sample standard deviation, n is the number of measurements, and t is a multiplier from the Student t distribution. For a 95% confidence interval, t depends on $n - 1$ degrees of freedom.

The quantity

$$\frac{s}{\sqrt{n}}$$

is called the **standard error of the mean**. It gets smaller when more independent measurements are taken. This does not mean systematic error disappears. It means the mean is better estimated relative to the random scatter.

n	3	4	5	6	10
Degrees of freedom	2	3	4	5	9
95% t value	4.303	3.182	2.776	2.571	2.262



The graph shows a critical idea: more independent measurements usually narrow the confidence interval, but the improvement becomes smaller and smaller. Taking 12 measurements instead of 3 helps, but it does not make the uncertainty vanish.

Worked example: A 95% confidence interval

A student standardizes a sodium hydroxide solution and obtains the following concentrations:

$$0.0996, \quad 0.1001, \quad 0.0998, \quad 0.1004, \quad 0.1000 \text{ mol dm}^{-3}.$$

For these data,

$$\bar{x} = 0.09998 \text{ mol dm}^{-3}, \quad s = 0.000303 \text{ mol dm}^{-3}, \quad n = 5.$$

For $n = 5$, there are 4 degrees of freedom, so $t = 2.776$ for a 95% confidence interval.

$$\text{CI half-width} = 2.776 \times \frac{0.000303}{\sqrt{5}} = 0.000377 \text{ mol dm}^{-3}.$$

So the result may be reported as

$$0.09998 \text{ mol dm}^{-3} \pm 0.00038 \text{ mol dm}^{-3} \quad (95\% \text{ CI}, n = 5).$$

Quick exercise: Confidence interval for concentration

Four independent trials give the following concentration values:

$$0.1012, \quad 0.1008, \quad 0.1010, \quad 0.1009 \text{ mol dm}^{-3}.$$

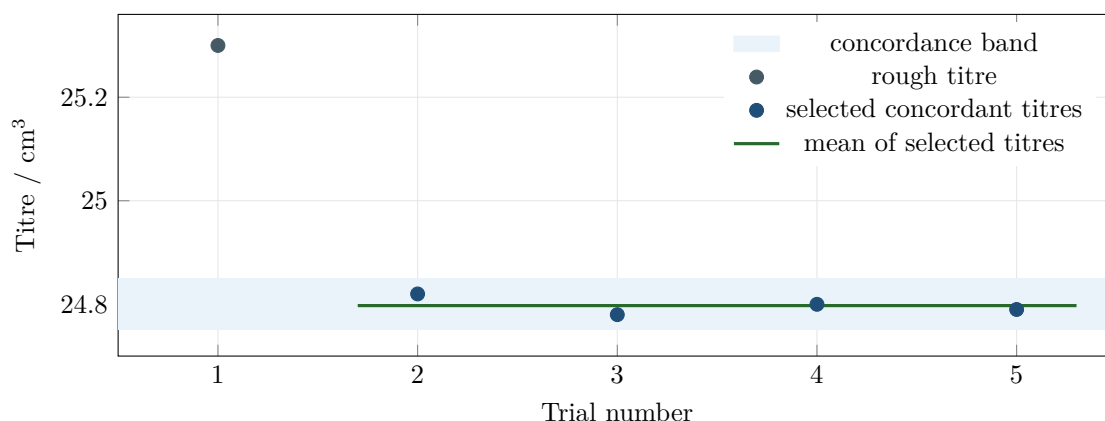
For $n = 4$, use $t = 3.182$ for a 95% confidence interval. Calculate the mean, sample standard deviation, and 95% confidence interval.

6 Concordant titres

In titration work, students often repeat titres until they obtain concordant results. A common school rule is that concordant titres should be within 0.10 cm^3 of each other, although some courses or teachers may use a stricter or looser criterion.

Concordance rule

A set of titres is concordant if the selected titres agree within a pre-defined range. The rule must be chosen before calculating the final mean. A rough titre is normally excluded from the final mean because it is used to locate the approximate endpoint.



In the plot, trial 1 is a rough titre. Trials 2–5 are concordant because they fall within a narrow band. The mean should be calculated from trials 2–5, not from all five values.

Worked example: Mean of concordant titres

The titres are:

Trial	1	2	3	4	5
Titre / cm³	25.30	24.82	24.78	24.80	24.79

Trial 1 is the rough titre. The mean of trials 2–5 is

$$\bar{x} = \frac{24.82 + 24.78 + 24.80 + 24.79}{4} = 24.7975 \text{ cm}^3 \approx 24.80 \text{ cm}^3.$$

Quick exercise: Choosing concordant titres

A student records the following titres:

Trial	1	2	3	4	5	6
Titre / cm³	24.95	24.61	24.63	24.62	24.64	24.88

Trial 1 was a rough titre. Use a concordance rule of 0.10 cm^3 .

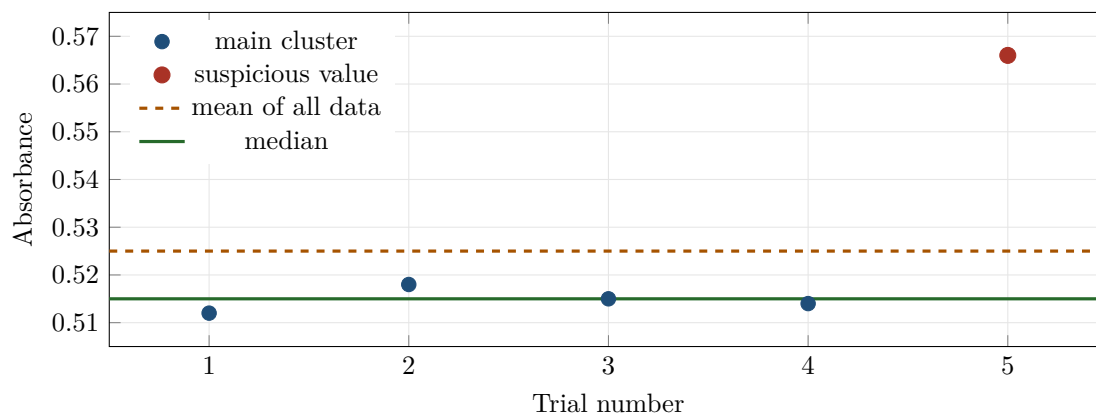
- Which trials should be used for the final mean?
- Calculate the final mean titre.
- How should trial 6 be described?

7 Outliers and data rejection

An **outlier** is a value that is unusually far from the rest of the data. Outliers matter because they can strongly affect the mean and standard deviation.

A strict rule for excluding data

Do not remove a value only because it makes the result look worse. Data rejection should be justified by a documented reason, a pre-defined rule, or a suitable statistical test. Otherwise, it becomes cherry-picking.



In the example above, the suspicious value pulls the mean upward. The median remains close to the main cluster. This does not prove that the value must be rejected, but it signals that the experimental record should be checked.

Practical decision sequence

When a value looks suspicious, use this sequence:

- 1) Check the lab record: Was there a spill, bubble, wrong endpoint, contaminated cuvette, or instrument warning?
- 2) Check whether a pre-defined rule applies, such as a concordance rule for titres.
- 3) Compare conclusions with and without the value.
- 4) Report the decision honestly.

Extension: Grubbs' test

In more advanced analytical chemistry, a formal outlier test such as Grubbs' test can be used for a single suspected outlier in approximately normally distributed data. The basic statistic is

$$G = \frac{|x_{\text{suspicious}} - \bar{x}|}{s}$$

The calculated G is compared with a critical value. This handout does not require Grubbs' test for routine calculations, but the key principle is important: the rule must be chosen before deciding whether to reject the value.

Quick exercise: Outlier decision

A student measures the absorbance of a sample five times:

0.512, 0.518, 0.515, 0.514, 0.566.

- a) Calculate the mean of all five values.
- b) Calculate the median.
- c) Explain why the final value is suspicious.
- d) State what extra evidence would justify excluding it.

8 Reporting repeated measurements

A final result should normally include the estimate, the uncertainty or scatter, the unit, and the number of repeated measurements.

Useful reporting formats

For raw repeatability:

$$\bar{x} \pm s \quad (n = \text{number of measurements}).$$

For uncertainty in the mean:

$$\bar{x} \pm 95\% \text{ CI} \quad (n = \text{number of independent measurements}).$$

Always state what the uncertainty represents. Standard deviation and confidence interval are not the same thing.

Weak reporting	Improved reporting
The titre was 24.80 cm ³ .	The mean concordant titre was 24.80 cm ³ from four titres, with $s = 0.017 \text{ cm}^3$.
The concentration was 0.100 mol dm ⁻³ .	The concentration was $0.09998 \text{ mol dm}^{-3} \pm 0.00038 \text{ mol dm}^{-3}$ (95% CI, $n = 5$).
One value was removed.	Trial 6 was excluded because a bubble was observed in the burette tip before the endpoint.

Quick exercise: Improve the reporting

Rewrite the following result as a stronger chemistry statement:

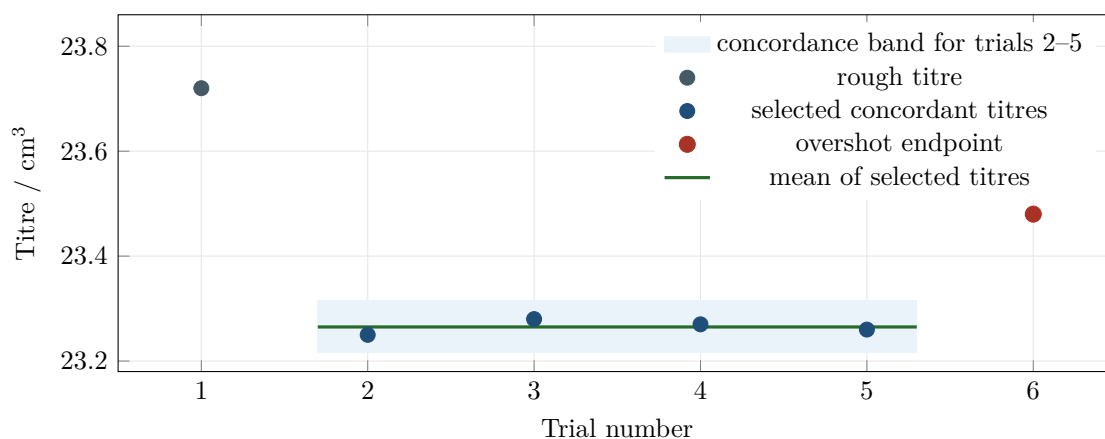
"We got 18.46 cm³, and one titre was bad, so we ignored it."

Use the terms **rough titre**, **concordant titres**, **mean**, and **standard deviation**.

9 Mini-practical data set

Use this data set to practise the full Block 2 workflow. A student titrates an acid sample with standardized sodium hydroxide.

Trial	Titre / cm ³	Notes
1	23.72	rough titre
2	23.25	endpoint sharp
3	23.28	endpoint sharp
4	23.27	endpoint sharp
5	23.26	endpoint sharp
6	23.48	endpoint overshoot; very pale pink persisted strongly



Quick exercise: Full Block 2 workflow

Use the mini-practical data set above.

- Identify the trials that should be used for the final result.
- Calculate the mean titre.
- Calculate the sample standard deviation.
- Calculate the RSD%.
- Calculate the 95% confidence interval using $t = 3.182$ for $n = 4$.
- Write the final reported titre in a scientifically acceptable sentence.

Answer key

Repeated readings or independent replicates. a) repeated readings; b) independent replicates; c) independent aliquot replicates, although they still depend on the same standardized NaOH solution.

Basic summary statistics. Data: 0.204, 0.207, 0.203, 0.206, 0.205. Mean = 0.205. Median = 0.205. Range = 0.207 – 0.203 = 0.004.

Standard deviation by calculation. Sample standard deviation $s = 0.00158 \approx 0.0016$. The absorbance readings typically scatter by about 0.0016 absorbance units around the mean.

RSD for repeated masses. Mean = 12.414 g. Sample standard deviation = 0.0207 g. RSD = 0.167% $\approx$ 0.17%.

Confidence interval for concentration. Mean = 0.100 975 mol dm⁻³. Sample standard deviation = 0.000 171 mol dm⁻³. 95% CI half-width:

$$3.182 \times \frac{0.000171}{\sqrt{4}} = 0.000\,272 \text{ mol dm}^{-3}.$$

A suitable report is

$$0.100\,98 \text{ mol dm}^{-3} \pm 0.000\,27 \text{ mol dm}^{-3} \quad (95\% \text{ CI}, n = 4).$$

Choosing concordant titres. Use trials 2–5. Their mean is

$$\frac{24.61 + 24.63 + 24.62 + 24.64}{4} = 24.625 \text{ cm}^3 \approx 24.63 \text{ cm}^3.$$

Trial 6 should be described as suspicious or non-concordant, not simply deleted without explanation. If there was endpoint overshoot or another recorded procedural issue, exclusion is stronger.

Outlier decision. Mean of all five values = 0.525. Median = 0.515. The value 0.566 is suspicious because it is far above the cluster 0.512–0.518 and pulls the mean upward. Exclusion would require extra evidence such as cuvette contamination, bubble, wrong wavelength, instrument warning, or a pre-defined statistical rejection rule.

Improve the reporting. Possible answer: “Trial 1 was treated as the rough titre. The concordant titres were trials 2–5. The final mean titre was calculated from the concordant titres only and reported with the sample standard deviation. Trial 6 was excluded because the endpoint was overshoot, as recorded in the notes.”

Full Block 2 workflow. Use trials 2–5. Mean:

$$\bar{x} = \frac{23.25 + 23.28 + 23.27 + 23.26}{4} = 23.265 \text{ cm}^3 \approx 23.27 \text{ cm}^3.$$

Sample standard deviation:

$$s = 0.0129 \text{ cm}^3 \approx 0.013 \text{ cm}^3.$$

RSD:

$$\frac{0.0129}{23.265} \times 100\% = 0.0555\% \approx 0.056\%.$$

95% CI half-width:

$$3.182 \times \frac{0.0129}{\sqrt{4}} = 0.0205 \text{ cm}^3 \approx 0.021 \text{ cm}^3.$$

Possible final report: “The mean concordant titre was $23.265 \text{ cm}^3 \pm 0.021 \text{ cm}^3$ (95% CI, $n = 4$), using trials 2–5. Trial 1 was the rough titre and trial 6 was excluded because the endpoint was overshoot.”